

Contemporary C++: *Learning Modern C++ in a Modern Way*



مدرس: سعید امراللہی بیوکی

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Agenda 22/30

Session 22. Introduction to C Memory model

- 📊 Fundamental types vs. User-defined types
- 📊 The C++ classes classification: Concrete and Handle classes
- 📊 Classes: Data members and Member functions
- 📊 References and further readings
- 📊 Q&A

120 min (incl. Q & A)





Memory management

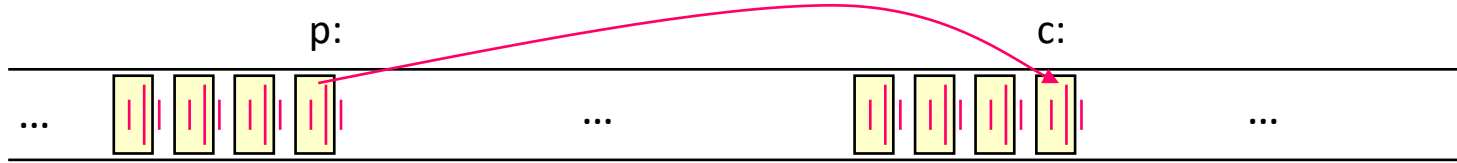
the C **M**emory model

- Programmer-controlled memory management



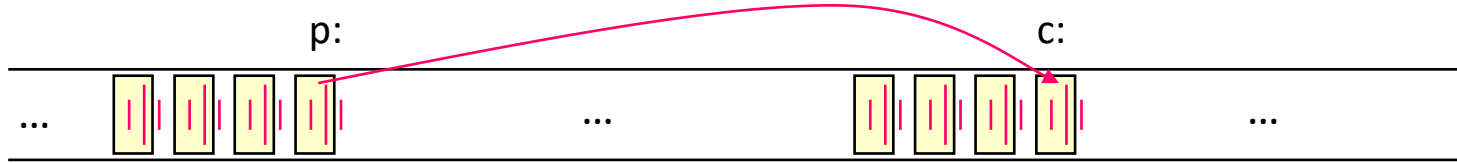
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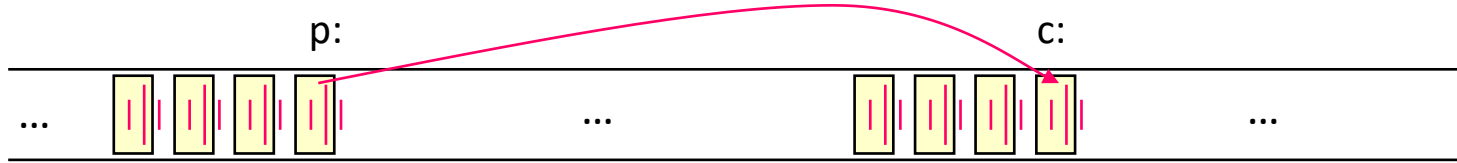


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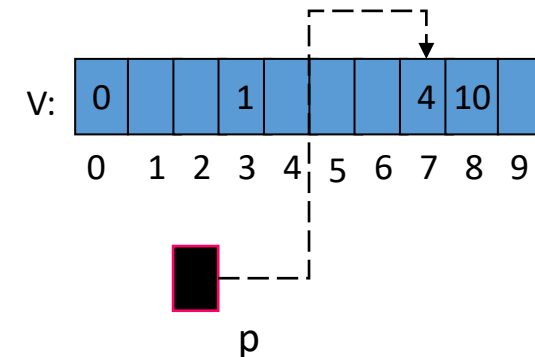
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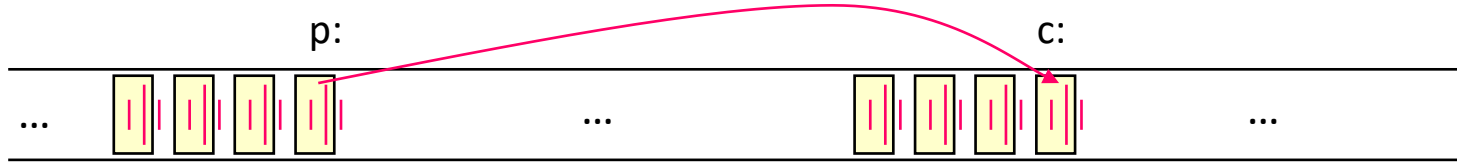
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int v[10]; // an array of 10 ints
int* p; // p is a pointer to an int
p = &v[7]; // assign the address of v[7] to p
*p = 4; // write to v[7] through p
*(p++) = 10;
*v = 0; // the name of array points to first element
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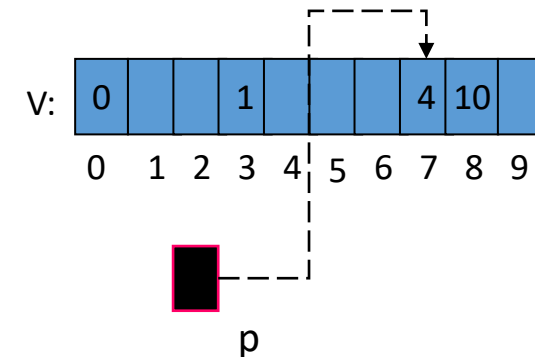
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- Pointer arithmetic
- No initialization, no bound checking



P rogrammer-controlled memory management

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- The concept of storage duration

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- **Free store**: Objects can be created dynamically during program execution using **new**, and destroyed using **delete**. The storage for these objects shall last for the duration of the program, or executing delete operator.
 - the keywords: **new** and **delete**
 - Free memory = dynamic memory = heap

R eferences and further readings

[Str13] Bjarne Stroustrup. The C++ Programming Language. 4th edition. Addison Wesley. Reading Mass. USA. May 2013.

Thanks for your patience ...

A man who asks a question is a fool for minute,
The man who does not ask, is a fool for a life.
- Confucius

Learning to ask the right (often hard) questions is an essential part of learning to think as a programmer.

- Bjarne Stroustrup *programming Principles and Practice Using C++, page 4.*

There is no stupid question, but there is stupid answer.
- Howard Hinnant

